STA for Nanometer Designs — 2 Week Plan

Book: Bhasker & Chadha — *Static Timing Analysis for Nanometer Designs* **Goal:** Job/internship onboarding — practical, Synopsys-flow-oriented **Budget:** ~2–3 hrs/day × 14 days **Coverage:** Chs 1–10 (pp. 1 ≈ 414), tied to Synopsys docs already in `Synopsys/`

Each day = a chapter slice + a Synopsys doc to skim alongside + a small output. The output is the part that actually transfers the knowledge into your head — don't skip it.

Week 1 — Foundations & Models

Day 1 · Ch 1 + Ch 2 §2.1–2.5 · *What STA is, propagation delay, slew, skew, timing arcs*

- **Read:** pp. 1–34 (~34p)
- **Synopsys:** `Fusion Compiler – User Guide.pdf` — skim front matter + STA flow overview
- **Output:** One-page sketch — block diagram of the STA flow (netlist + .lib + SDC + SPEF → PrimeTime → reports)

Day 2 · Ch 2 §2.6–2.10 · *Min/max paths, clock domains, operating conditions (PVT)*

- **Read:** pp. 34–42 (~9p) — short day, use the slack for review
- **Synopsys:** `Fusion Compiler – Timing Analysis.pdf` — intro chapters only
- **Output:** Note the 4 path types (in→reg, reg→reg, reg→out, in→out) and which corner you'd run each at

Day 3 · Ch 3 §3.1–3.4 · *Pin cap, NLDM, combinational + sequential cell models*

- **Read:** pp. 43–68 (~26p) — densest chapter, take your time
- **Synopsys:** `Fusion Compiler – Data Model.pdf` — focus on .lib structure
- **Output:** Hand-sketch the NLDM 2D lookup table (input slew × output cap → delay). Add setup/hold check definitions in your notes.

Day 4 · Ch 3 §3.5–3.10 · *State-dependent, receiver pin cap, noise models, power, derating*

- **Read:** pp. 70–100 (~30p)
- **Synopsys:** `Fusion Compiler – Power Analysis.pdf` — skim Active vs Leakage power sections
- **Output:** Write out the k-factor derating formula and what "best/typical/worst case" means in each PVT corner

Day 5 · Ch 4 · *Interconnect Parasitics*

- **Read:** pp. 101–122 (~22p)
- **Synopsys:** `Fusion Compiler – Timing Analysis.pdf` — SPEF / DSPF section
- **Output:** Side-by-side sketch of T-model vs Pi-model. Note which one PrimeTime uses and why.

Day 6 · Ch 5 · *Delay Calculation*

- **Read:** pp. 123–146 (~24p) — this is the math-heavy core
- **Synopsys:** Re-skim `Fusion Compiler – Timing Analysis.pdf` (delay calc section)
- **Output:** Do one Elmore delay hand calculation on a 3-stage RC chain. Then write 2 lines on what "effective capacitance" buys you over total capacitance.

Day 7 · Ch 6 · *Crosstalk and Noise*

- **Read:** pp. 147–178 (~32p)
- **Synopsys:** `Fusion Compiler – Error Messages.pdf` — grep for "noise" and "crosstalk" related messages
- **Output:** Aggressor/victim diagram. Write the formula for crosstalk delta delay and explain when delta is positive vs negative.

Weekend check-in: You should be comfortable explaining *what* STA does and *how delays/parasitics are modeled*. If a section felt foggy, this is the day to revisit it before constraints week.

Week 2 — Constraints, Verification, Interfaces

Day 8 · Ch 7 §7.1–7.6 · *Clocks, generated clocks, latency, jitter, uncertainty*

- **Read:** pp. 179–204 (~26p)
- **Synopsys:** `Fusion Compiler – Verilog.pdf` / `SystemVerilog.pdf` — RTL conventions
- **Output:** Write a real SDC snippet:

  ```
  create_clock -name CLK -period 10 [get_ports clk]
  create_generated_clock -name CLK_DIV2 -source [get_ports clk] -divide_by 2 [get_pi
  set_clock_latency 0.5 [get_clocks CLK]
  set_clock_uncertainty 0.2 [get_clocks CLK]
  ```

Day 9 · Ch 7 §7.7–7.12 · *I/O delays, false paths, multicycle, case analysis, mode disable*

- **Read:** pp. 204–226 (~23p)
- **Synopsys:** `Fusion Compiler – Tool Commands.pdf` — look up `set_false_path`, `set_multicycle_path`, `set_case_analysis`

- **Output:** Write a full SDC for a fake module — clocks, I/O delays, one false path, one multicycle. ~15 lines.

Day 10 · Ch 8 §8.1–8.3 · ***Setup/hold checks, async checks, recovery/removal***

- **Read:** pp. 227–272 (~46p) — big chapter, split is intentional
- **Synopsys:** `Fusion Compiler – Timing Analysis.pdf` — report sections
- **Output:** Annotated waveform showing a setup check passing and one failing. Write the setup equation: `Tclk ≥ Tclk-q + Tcomb + Tsetup − Tskew`

Day 11 · Ch 8 §8.4–8.6 · ***Timing reports, debug flow, examples***

- **Read:** pp. 272–316 (~45p)
- **Synopsys:** `Fusion Compiler – Tool Commands.pdf` — `report_timing` and its options
- **Output:** Memorize the fields of a `report_timing` output (launch clock, capture clock, data path, slack). Write a fake one from scratch — no peeking.

Day 12 · Ch 9 §9.1–9.3 · ***Async interfaces, source-synchronous basics***

- **Read:** pp. 317–340 (~24p)
- **Synopsys:** `Fusion Compiler – User Guide.pdf` — I/O interface section
- **Output:** Source-synchronous clock-data timing diagram. Label tCO, setup, hold windows relative to the launched clock.

Day 13 · Ch 9 §9.4–end · ***DDR interfaces, centered vs edge-aligned***

- **Read:** pp. 340–364 (~25p)
- **Synopsys:** `Fusion Compiler – Multivoltage.pdf` — skim level-shifter timing
- **Output:** Write the setup/hold equations for a DDR interface. Sketch edge-aligned vs centered DQ/DQS.

Day 14 · Ch 10 · ***Robust Verification + final review***

- **Read:** pp. 365–~414 (~50p)
- **Synopsys:** `Fusion Compiler – Design for Test.pdf` — clock gating in scan
- **Output:** Two artifacts:
  1. Clock gating check example (setup + hold gating)
  2. **1-page personal cheat sheet** — all the equations, the SDC commands, the report fields. This is what you'll re-read before an interview or your first day.

Daily routine (keep it tight, 2–3 hrs)

1. **~75 min** — read the assigned pages, highlight equations and definitions
2. **~30 min** — open the paired Synopsys doc, skim relevant section

3. **~30 min** — do the day's output (sketch / SDC / hand calc)
4. **~5 min** — write 3 lines in a notes file: what was new, what was confusing, what to revisit

If you fall behind

- Skip the Synopsys doc skim before skipping the reading
- The "must-not-skip" days are **3, 6, 8, 10, 14** — these are the load-bearing chapters
- Chapter 9 (interfaces) is interview-favorite; don't compress it

Reference companions in this folder

- `Synopsys/Fusion Compiler – Timing Analysis.pdf` — your main flow doc
- `Synopsys/Fusion Compiler – Tool Commands.pdf` — SDC command reference
- `Verilog & SystemVerilog/IEEE 1800 – SystemVerilog LRM.pdf` — for any RTL question that comes up